Probability & Statistics

A concise formula sheet | Counting, probability, descriptive statistics and random variables

1. Counting Principles

Multiplication Rule

If a process has r successive stages, with n_i choices at stage i, then the total number of outcomes is

$$n_1 n_2 \cdots n_r.$$

Permutations: order matters

The number of ways to choose and arrange k objects from n distinct objects:

$$ {}_nP_k = \frac{n!}{(n-k)!}.$$

All n objects arranged: $n!$
Circular permutations: $(n-1)!$

Combinations: order does not matter

The number of ways to choose k objects from n distinct objects:

$$ {}_nC_k = \binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Permutations with indistinguishable objects

If there are n objects, with $n_1, n_2, \ldots, n_k$ identical objects of each type, where $\sum_i n_i = n$, then

$$\frac{n!}{n_1! n_2! \cdots n_k!}.$$

2. Set Algebra

Set notation and identities

- $A \cup B$: union (or); $A \cap B$: intersection (and).
- A' or A^c: complement; S: sample space.
- $A \cap A' = \varnothing, \quad A \cup A' = S.$
- $A \cup \varnothing = A, \quad A \cap \varnothing = \varnothing.$
- $A \cup S = S, \quad A \cap S = A.$
- $A - B = A \cap B'.$

De Morgan's laws

$$(A \cup B)' = A' \cap B', \qquad (A \cap B)' = A' \cup B'.$$

Distributive laws

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$
$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

3. Probability Rules

Axioms and complements

$$\mathbb{P}(S) = 1, \qquad 0 \le \mathbb{P}(A) \le 1$$
$$\mathbb{P}(A') = 1 - \mathbb{P}(A).$$

If A and A' are complements, exactly one occurs.

Addition Rule

For any two events:

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

If A and B are mutually exclusive:

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B).$$

For three events:

$$\begin{aligned}\mathbb{P}(A \cup B \cup C) = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) \\ - \mathbb{P}(A \cap B) - \mathbb{P}(A \cap C) \\ - \mathbb{P}(B \cap C) + \mathbb{P}(A \cap B \cap C).\end{aligned}$$

Mutually Exclusive Events

Events are mutually exclusive if they cannot occur together:

$$A \cap B = \varnothing, \qquad \mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B).$$

Conditional Probability

The probability of A, given that B has occurred:

$$\boxed{\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}} \quad \text{if } \mathbb{P}(B) > 0.$$

Multiplication rule:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\mathbb{P}(B).$$

Independence

Events A and B are independent if knowing that one occurred does not change the probability of the other:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

Equivalently, if $\mathbb{P}(B) > 0$:

$$\mathbb{P}(A \mid B) = \mathbb{P}(A).$$

Mutually exclusive events with positive probabilities are not independent.

Bayes' Theorem

For a partition $A_1, \ldots, A_k$ of the sample space:

$$\mathbb{P}(A_i \mid B) = \frac{\mathbb{P}(B \mid A_i)\mathbb{P}(A_i)}{\sum_{j=1}^{k} \mathbb{P}(B \mid A_j)\mathbb{P}(A_j)}.$$

For two events:

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A)\mathbb{P}(A)}{\mathbb{P}(B)}.$$

4. Descriptive Statistics

Measures of Location

Mean (average)

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}.$$

For a random variable, the population mean is $\mu = \mathbb{E}[X]$.
Median: middle value after sorting the data. For even n, average the two middle values.
Mode: most frequently occurring value.
Quartiles: Q_1 (25th percentile), Q_2 (median, 50th), Q_3 (75th).
Percentile position (one common convention): sort the data and use position $L = (n+1)p$, where p is the percentile expressed as a proportion. Interpolate if L is not an integer. Conventions can differ between calculators and textbooks.

Measures of Variability

Range

$$\max(x_i) - \min(x_i).$$

Interquartile range

$$IQR = Q_3 - Q_1.$$

Mean absolute deviation about the mean

$$MAD = \frac{1}{n}\sum_{i=1}^{n} |x_i - \bar{x}|.$$

Population variance and standard deviation

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2, \qquad \sigma = \sqrt{\sigma^2}.$$

Sample variance and standard deviation

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2, \qquad s = \sqrt{s^2}.$$

5. Random Variables

Definitions

A random variable X assigns a real number to each outcome of a random experiment.
Discrete: takes a finite or countable set of values.
Continuous: takes values over intervals of the real line.

Probability Distributions

A valid probability distribution satisfies:

$$p(x) \geq 0, \qquad \sum_{x} p(x) = 1$$

for a discrete random variable, and

$$f(x) \geq 0, \qquad \int_{-\infty}^{\infty} f(x)\,dx = 1$$

for a continuous random variable.

6. Expectation and Variance

Expected Value

For a discrete random variable:

$$\mathbb{E}[X] = \sum_x x\, p(x).$$

For a continuous random variable:

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f(x)\, dx.$$

For a function $g(X)$:

$$\mathbb{E}[g(X)] = \sum_x g(x)p(x) \quad \text{or} \quad \int_{-\infty}^{\infty} g(x)f(x)\, dx.$$

Variance and Standard Deviation

$$\mathrm{Var}(X) = \mathbb{E}[(X-\mu)^2] = \mathbb{E}[X^2] - \mu^2.$$

$$\sigma_X = \sqrt{\mathrm{Var}(X)}.$$

For a discrete variable:

$$\mathrm{Var}(X) = \sum_x (x-\mu)^2 p(x).$$

For a continuous variable:

$$\mathrm{Var}(X) = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\, dx.$$

Linearity and Transformations

$$\mathbb{E}[aX+b] = a\mathbb{E}[X] + b.$$

$$\mathrm{Var}(aX+b) = a^2\,\mathrm{Var}(X).$$

More generally:

$$\mathbb{E}[g(X)+h(X)] = \mathbb{E}[g(X)] + \mathbb{E}[h(X)].$$

Two Random Variables

If X and Y are independent:

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y], \qquad \mathrm{Cov}(X,Y) = 0.$$

For any a, b:

$$\mathrm{Var}(aX+bY) = a^2\,\mathrm{Var}(X) + b^2\,\mathrm{Var}(Y)$$

when X, Y are independent. In general:

$$\mathrm{Var}(aX+bY) = a^2\,\mathrm{Var}(X) + b^2\,\mathrm{Var}(Y) + 2ab\,\mathrm{Cov}(X,Y).$$

7. Discrete Probability Distributions

Discrete Uniform Distribution

If X has n equally likely possible values:

$$\mathbb{P}(X = x_i) = \frac{1}{n}.$$

For consecutive integers $a, a+1, \ldots, b$:

$$\mathbb{E}[X] = \frac{a+b}{2}, \qquad \mathrm{Var}(X) = \frac{(b-a+1)^2 - 1}{12}.$$

Bernoulli Distribution

One trial with two outcomes: success (1) with probability p, failure (0) with probability $1-p$.

$$\mathbb{P}(X = x) = p^x (1-p)^{1-x}, \quad x \in \{0, 1\}.$$

$$\mathbb{E}[X] = p, \qquad \mathrm{Var}(X) = p(1-p).$$

Binomial Distribution

Counts the number of successes in n independent Bernoulli trials, each with success probability p.

$$X \sim \mathrm{Bin}(n, p)$$

$$\mathbb{P}(X = x) = \binom{n}{x} p^x (1-p)^{n-x}.$$

$$\mathbb{E}[X] = np, \qquad \mathrm{Var}(X) = np(1-p).$$

Geometric Distribution

Number of trials until the first success, with independent trials and success probability p:

$$\mathbb{P}(X = k) = (1-p)^{k-1} p, \quad k = 1, 2, \ldots$$

$$\mathbb{E}[X] = \frac{1}{p}, \qquad \mathrm{Var}(X) = \frac{1-p}{p^2}.$$

Memoryless property:

$$\mathbb{P}(X > s + t \mid X > s) = \mathbb{P}(X > t).$$

Poisson Distribution

Models counts of events in a fixed interval when events occur independently at a constant average rate.

$$X \sim \mathrm{Pois}(\lambda), \qquad \mathbb{P}(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}.$$

For rate r per unit time over interval length t, $\lambda = rt$:

$$\mathbb{E}[X] = \lambda, \qquad \mathrm{Var}(X) = \lambda.$$

Poisson can approximate a binomial distribution when n is large and p is small, with $\lambda = np$.

8. Continuous Probability Distributions

Continuous Uniform Distribution

For $X \sim U(a, b)$:

$$f(x) = \begin{cases} \dfrac{1}{b-a}, & a \le x \le b, \\ 0, & \text{otherwise.} \end{cases}$$

$$\mathbb{E}[X] = \frac{a+b}{2}, \qquad \mathrm{Var}(X) = \frac{(b-a)^2}{12}.$$

Normal Distribution

A bell-shaped, symmetric distribution determined by mean μ and standard deviation σ:

$$X \sim N(\mu, \sigma^2).$$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2} \right).$$

Mean = median = mode = μ. The parameter σ controls spread.

Standard Normal Distribution

Standardize a normal variable using:

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1).$$

If $\Phi(z) = \mathbb{P}(Z \le z)$, then:

$$\mathbb{P}(a < Z < b) = \Phi(b) - \Phi(a).$$

$$\mathbb{P}(Z > z) = 1 - \Phi(z), \qquad \mathbb{P}(Z < -z) = 1 - \Phi(z).$$

For a symmetric interval:

$$\mathbb{P}(-z < Z < z) = 2\Phi(z) - 1.$$

Use a standard normal table or calculator to find $\Phi(z)$.

Exponential Distribution

Models waiting time between events in a Poisson process with rate λ:

$$f(x) = \lambda e^{-\lambda x}, \quad x \ge 0.$$

$$\mathbb{E}[X] = \frac{1}{\lambda}, \qquad \mathrm{Var}(X) = \frac{1}{\lambda^2}.$$

Cumulative distribution function:

$$F(x) = \mathbb{P}(X \le x) = 1 - e^{-\lambda x}, \quad x \ge 0.$$

9. Cumulative and Joint Distributions

PMF, PDF and CDF

Discrete PMF

$$p_X(x) = \mathbb{P}(X = x).$$

Continuous PDF

$$\mathbb{P}(a \le X \le b) = \int_a^b f_X(x)\, dx.$$

CDF (both cases)

$$F_X(x) = \mathbb{P}(X \le x).$$

For a continuous variable:

$$F_X(x) = \int_{-\infty}^{x} f_X(t)\,dt, \qquad f_X(x) = F_X'(x)$$

where the derivative exists. A CDF is non-decreasing and ranges from 0 to 1.

Joint and Marginal Distributions

For discrete random variables:

$$p_{X,Y}(x,y) = \mathbb{P}(X = x, Y = y).$$

$$p_X(x) = \sum_y p_{X,Y}(x,y), \qquad p_Y(y) = \sum_x p_{X,Y}(x,y).$$

For continuous random variables:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dx.$$

The joint density integrates to 1 over its full support.

Conditional Distributions

For discrete variables:

$$p_{X|Y}(x|y) = \frac{p_{X,Y}(x,y)}{p_Y(y)}.$$

For continuous variables:

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}.$$

Independence of Random Variables

X and Y are independent if their joint distribution factors:

$$p_{X,Y}(x,y) = p_X(x)p_Y(y)$$

for discrete variables, or

$$f_{X,Y}(x,y) = f_X(x)f_Y(y)$$

for continuous variables.

10. Sampling and Limit Theorems

Central Limit Theorem (CLT)

For a sufficiently large sample of independent, identically distributed observations with population mean μ and finite variance σ^2, the sample mean is approximately normal:

$$\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right).$$

Standard error:

$$SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

Standardized sample mean:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}.$$

Properties of the Sample Mean

For independent observations from a population with mean μ and variance σ^2:

$$\mathbb{E}[\bar{X}] = \mu, \qquad \mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n}.$$

As sample size n increases, the standard error decreases.

Difference Between Two Independent Sample Means

For independent samples:

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}.$$

This is standard normal when the population variances are known and the sampling distribution is normal (or approximately normal under suitable CLT conditions).

Normal Approximation to the Binomial

For $X \sim \mathrm{Bin}(n,p)$, use a normal approximation when np and $n(1-p)$ are sufficiently large (a common rule is both at least 5 or 10):

$$X \approx N(np, np(1-p)).$$

With continuity correction:

$$\mathbb{P}(X \le x) \approx \mathbb{P}\left(Y \le x + \frac{1}{2}\right),$$

where $Y \sim N(np, np(1-p))$. Thus

$$z = \frac{x + 0.5 - np}{\sqrt{np(1-p)}}.$$

Linear Interpolation

If (x_1, p_1) and (x_2, p_2) are known points on a straight line, then for $x_1 < x < x_2$:

$$\frac{x - x_1}{x_2 - x_1} = \frac{p - p_1}{p_2 - p_1}.$$